4

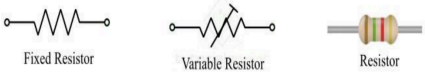
What is an Electric Circuit?

An Electric Circuit is a closed path for transmitting an electric current through the medium of electrical and magnetic fields. The flow of electrons across the loop constitutes the electric current.

Basic Circuit Elements:

Active Circuit Elements: Active Elements are those which can generate energy. Examples include batteries, generators, operational amplifiers and diodes. Note that in an electrical circuit, the source elements are the most significant active elements.

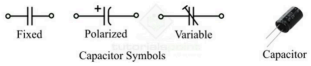
Resistor: A resistor opposes the flow of current through it.



Inductor: An inductor stores energy in form of the electromagnetic field.



Capacitor: A capacitor stores energy in form of the electrostatic field. The voltage across a capacitor is proportional to the charge.



5

Kirchhoff's Law

Kirchhoff's Current Law

Kirchhoff's Current Law states that" the algebraic sum of all the currents at any node point or a junction of a circuit is zero". $\Sigma I = 0$

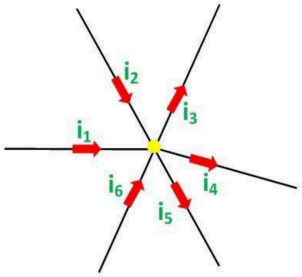
Considering the above figure as per the Kirchhoff's Current Law:

$i_1 + i_2 - i_3 - i_4 - i_5 + i_6 = 0$ ..... (1)

The direction of incoming currents to a node is taken as positive while the outgoing currents are taken as negative. The reverse of this can also be takenThe equation (1) can also be written as:

$i_1 + i_2 + i_6 = i_3 + i_4 + i_5$

Sum of incoming currents = Sum of outgoing currents



According to the **Kirchhoff's Current Law**, the algebraic sum of the currents entering a node must be equal to the algebraic sum of the currents leaving the node in an electrical network.

Considering the other figure shown below and assuming the direction of the current i

6

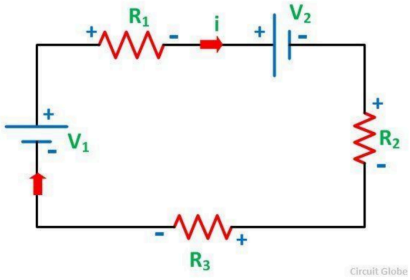
Kirchhoff's Voltage Law

Kirchhoff's Voltage Law states that the algebraic sum of the voltages (or voltage drops) in any closed path of a network that is transverse in a single direction is zero. In other words, in a closed circuit, the algebraic sum of all the EMFs and the algebraic sum of all the voltage drops (product of current (I) and resistance (R)) is zero.

$$-V_1 + iR_1 + V_2 + iR_2 + iR_3 = 0 \ldots\ldots\ldots (3) \quad \text{or}$$

$$i(R_1 + R_2 + R_3) = V_1 - V_2 \quad \text{or}$$

$$i = \frac{V_1 - V_2}{R_1 + R_2 + R_3}$$



It is seen that the voltage V_1 is negative and V_2 is positive
For the dependent sources in the circuit, KVL can also be applied.